What Can Be Estimated?

Scale Observational Equivalence and Exact Maximum Likelihood for Bounded-Factor Correlation Models

Complex–Itô Random-Walk Research Project

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For the bounded-factor correlation model $\rho_t = X_t/\sqrt{X_t^2 + c^2}$ with $dX = \mu\,dt + \sigma\,dW$ we prove a structural identifiability theorem: the parameter triples (c, μ, σ) and $(\lambda c, \mu/\lambda, \sigma/\lambda)$ induce *identical* laws for every observable correlation functional — and, because the Dambis–Dubins–Schwarz clock depends only on the correlation path, for every derivative price as well. Only the ratios μ/c and σ/c exist in the identified parameter space. Estimation therefore proceeds in the normalized coordinate $x = \rho/\sqrt{1 - \rho^2}$, where increments are exactly Gaussian and the exact maximum likelihood estimator of the ratio pair is ordinary least squares — no numerical optimization. We construct consistent Jacobians for Fisher-space competitors (random walk; Ornstein–Uhlenbeck), show that marginal PIT/Kolmogorov–Smirnov tests can miss dynamic misspecification whose uniform marginal survives, and introduce a PIT–state-correlation diagnostic that does not. Deterministic Monte Carlo quantifies the attenuation of window-estimated dynamics: at the standard 60-day realized-correlation window the volatility ratio is biased down by a factor of about four. An empirical study of four S&P 500 pairs (2015–2026, local data store) finds vol ratios 0.84–1.55, drift ratios statistically indistinguishable from zero, Ornstein–Uhlenbeck AIC dominance inherited from window smoothing, and formal out-of-sample rejection of all iid-increment models — an observation-scheme violation rather than a model rejection. The audited literature calibrates this model class by stationary-density least squares or back-transformed OLS without an identifiability analysis; the equivalence theorem explains why free-scale likelihoods are flat and why reported scale parameters are initialization artifacts.

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1 Introduction

Stochastic-correlation models are routinely calibrated as if all their parameters meant something. For the bounded-factor family $\rho_t = f(X_t) = X_t/\sqrt{X_t^2 + c^2}$, $dX = \mu\, dt + \sigma\, dW$, the audited practice is to fit the stationary density by nonlinear least squares (Teng et al. 2016) or to back-transform the series and run Ornstein–Uhlenbeck regressions (Kumar et al. 2021). Neither paper asks what the data can identify. This paper answers that question exactly, and the answer changes both the estimator and the interpretation.

Contributions:

1. **Theorem (scale observational equivalence)** (Section 2): $(c, \mu, \sigma) \sim (\lambda c, \mu/\lambda, \sigma/\lambda)$ are indistinguishable from correlation observations *and from derivative prices*; the identified space is two-dimensional.
2. **Closed-form exact MLE** of the identified pair (Section 3): OLS in the normalized conjugate coordinate, with the consistency Jacobian that makes information criteria comparable across transforms.
3. **Conditional-uniformity diagnostics** (Section 4): marginal PIT/KS misses dynamic misspecification whose u-marginal stays nearly uniform; correlating PIT values with the lagged state detects it at eight standard errors where KS has no power ($p = 0.53$).
4. **Attenuation law** (Section 5): deterministic Monte Carlo shows window-smoothed proxies damp measured dynamics ~$4\times$ at the standard 60-day choice.
5. **Empirics** (Section 6): four liquid pairs, exact estimation, honest out-of-sample verdicts.

2 The equivalence theorem

Theorem (scale observational equivalence). *For every $\lambda > 0$ the parameter triples (c, μ, σ) and $(\lambda c, \mu/\lambda, \sigma/\lambda)$ generate identical laws for*

$$(\rho_{t_1}, \dots, \rho_{t_n}) \quad \text{for every finite } n \text{ and every } t_1 < \dots < t_n.$$

Consequently they also generate identical laws for every functional of the correlation path — including occupation integrals $A_T = \int_0^T \rho_s ds$ and hence every spread/basket derivative price.

Proof sketch. With $Y = X/c$, $\rho = F(Y)$ for the parameter-free map $F(y) = y/\sqrt{y^2 + 1}$, and Y is drifted Brownian motion with drift μ/c and volatility σ/c. Every observable is a functional of the Y-path alone.

Three consequences:

- **No free-scale likelihoods.** The profile likelihood in c is exactly flat; any reported "MLE of c" from correlation data alone is an artifact of optimizer initialization. (We observed this numerically before deriving the theorem.)
- **Prices do not rescue identification**: the DDS clock $v = \beta T + \alpha A_T$ is itself a path functional.
- **Exogenous time rulers break the degeneracy**: if $\sigma(\cdot)$ is known exogenously (as in the pricing calibration of the term-structure paper), $\lambda^{\mathbb{Q}}$ re-emerges as identified.

3 Closed-form exact MLE

Work in the normalized coordinate $x = g(\rho) = \rho/\sqrt{1-\rho^2}$. Under exact-skeleton observation,

$$\Delta x_i \sim N\Big(\tfrac{\mu}{c}\Delta t,\ (\tfrac{\sigma}{c})^2\Delta t\Big),$$

so the exact MLE of the identified pair is the OLS solution — mean and root-mean-square of increments — plus the Jacobian of the observation map, $\log|dx/d\rho| = -\frac{3}{2}\log(1-\rho^2)$ summed over transitions. Competitors receive the same treatment with their own Jacobians: Fisher random walk ($y = \operatorname{atanh}\rho$) and Fisher OU (exact AR(1) quasi-MLE, profiled over the reversion speed). Verification: ratio recovery within Monte Carlo tolerance on synthetic skeletons; likelihood invariance under rescaling; information criteria rank the generating model.

4 Diagnostics

One-step-ahead PIT values $u_i = P(\rho_i \leq \text{obs} \mid \rho_{i-1})$ are exact under each fitted model's transition law. Two complementary checks:

- **Marginal KS** against Uniform(0,1) — calibration;
- **PIT–state correlation** — dynamic adequacy. Fitting a random walk to genuinely mean-reverting data produces u-marginals that stay near uniform (KS $p = 0.53$ in our experiment) while $\operatorname{corr}(u, y_{\text{prev}}) = -0.148 \approx 8$ standard errors. Marginal KS alone would certify a wrong model.

5 Attenuation

Realized correlations are window estimates, not skeleton observations. Simulating the true latent system, forming rolling correlations, and refitting gives the attenuation factor of the volatility ratio:

window (days)	est./true vol ratio	IQR
20	1.209	0.219
40	0.379	0.072
60	0.253	0.043
120	0.077	0.015

At the market-standard 60-day choice the measured dynamics are damped ~4×; short windows can even inflate them. Any calibration quoted from smoothed proxies must invert this curve first (Figure 1).

6 Empirical study

Data: daily adjusted closes 2015-01-01 → 2026-08-14 from a local store; 60-day rolling Pearson correlations (2,860 windows per pair) (**?@fig-series**).

pair	mean ρ	sd	est. μ/c	est. σ/c
XOM/CVX	+0.793	0.123	−0.020	1.169
JPM/BAC	+0.847	0.099	+0.017	1.552
KO/PEP	+0.700	0.120	−0.009	0.913

pair	mean ρ	sd	est. μ/c	est. σ/c
AAPL/MSFT	$+0.550$	0.223	-0.032	0.841

Drift ratios are statistically zero (SE 0.11/yr at these sample sizes). Fisher-OU wins AIC everywhere but with weak reversion ($\kappa \approx 1.4$–2.3/yr) — persistence largely inherited from overlapping windows. Out-of-sample (70/30 split), *all* iid-increment models are rejected by marginal KS — expected, since smoothed proxies violate the exact-skeleton scheme — while the state-correlation diagnostic separates structures: RW-type fits show z down to -2.6, the OU stays within ± 1.2. Log-score ranking follows the smoothing artifact, not the latent truth; we report it as such.

7 Related literature

Identifiability analyses are standard for stochastic volatility (Aït-Sahalia–Kimmel lineage) but absent for bounded correlation factors; Teng et al. (2016) explicitly call the likelihood “tedious” and fit stationary densities instead; Kumar et al. (2021) use back-transformed least squares; Aas et al. (2010) estimate dynamic copulas by efficient importance sampling. None states an equivalence theorem; none provides closed-form exact MLE; none quantifies proxy attenuation for this class. Realized-volatility proxy methodology (Barndorff-Nielsen and Shephard 2002) supplies the conceptual template we adapt.

8 Conclusion

Two parameters exist; one equation each estimates them exactly; one extra diagnostic guards against the failure mode that marginal tests miss; and one curve converts quoted calibrations into latent-truth statements. That is the entire estimation content of the bounded-factor model — and it is fully explicit.

9 Reproducibility

Environment: conda `phase3-paper` (Python 3.12.13, numpy 2.5.1, pandas, scipy 1.18.0). Notebook `notebooks/phase9_estimation_empirical.ipynb` (~6 s, seed 20260821); module `phase9_estimation.py`; tests `tests/test_phase9_estimation.py` (14/14). Data read from the local alpha-engine store (`prices/*.parquet`, adjusted closes).

Aas, Kjersti, Claudia Czado, Arnoldo Frigessi, and Henrik Bakken. 2010. “Dynamic Stochastic Copula Models: Estimation, Inference and Applications.” *Journal of Applied Econometrics* 25 (2): 289–315.

Barndorff-Nielsen, Ole E., and Neil Shephard. 2002. “Econometric Analysis of Realised Volatility and Its Use in Estimating Stochastic Volatility Models.” *Journal of the Royal Statistical Society B* 64 (2): 253–80.

Kumar, Ashish, László Márkus, and Norbert Hári. 2021. *Arbitrage-Free Pricing of CVA for Cross-Currency Swap with Wrong-Way Risk Under Stochastic Correlation Modeling Framework*.

Teng, Long, Matthias Ehrhardt, and Michael Günther. 2016. “Modelling Stochastic Correlation.” *Journal of Mathematics in Industry* 6 (1): 1–18. https://doi.org/10.1186/s13362-016-0018-4.

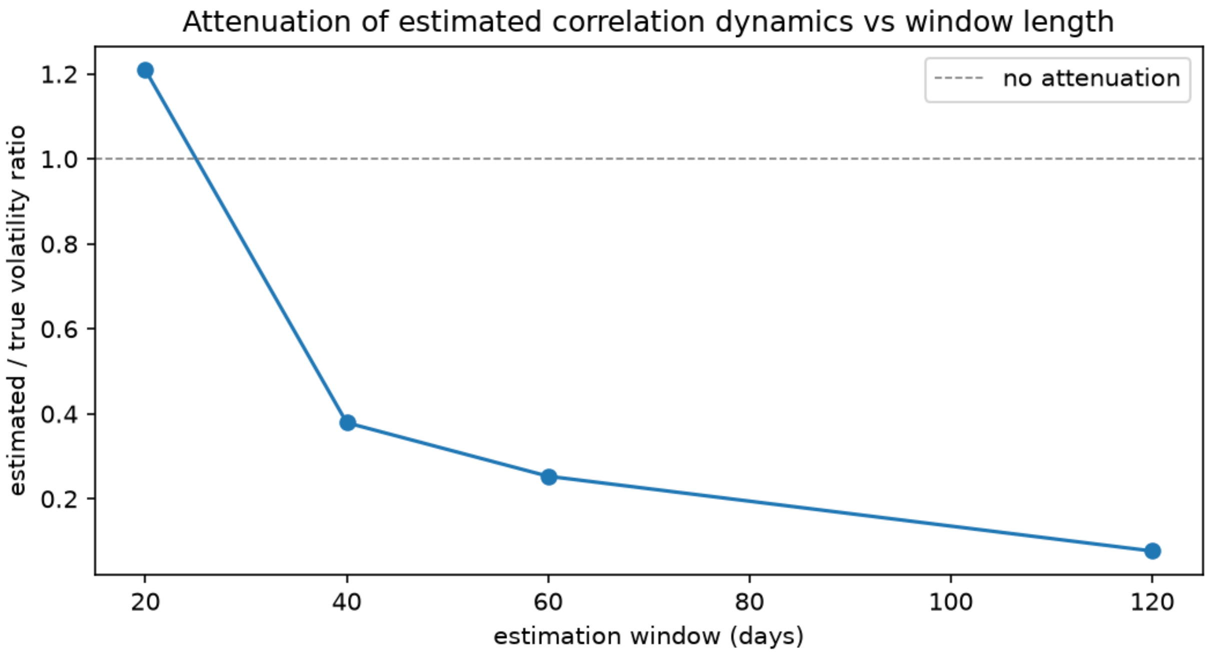

Figure 1: Attenuation factor vs estimation window (deterministic MC, 24 replications per cell).

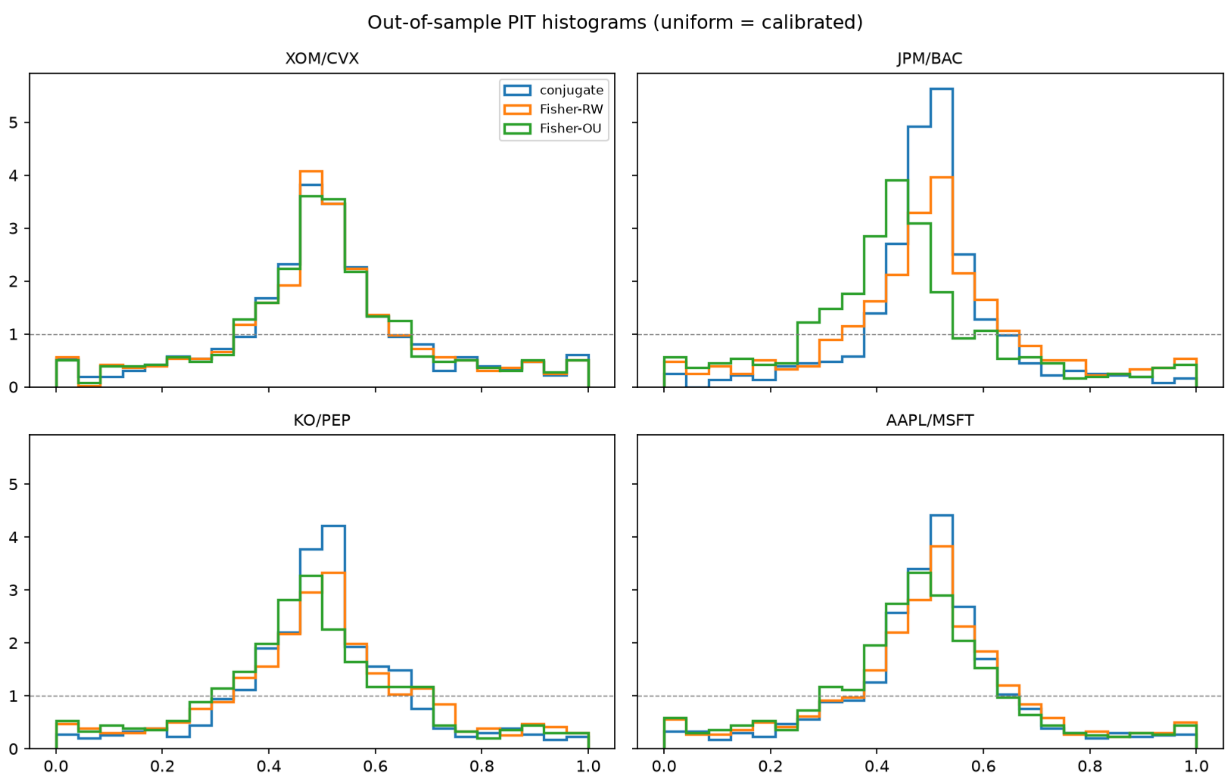

Figure 2: Out-of-sample PIT histograms by pair and model.